

Thank you!

[← Back to Presentations](#)



OmapSketch

Characterise databases

2026-09-17

Overview and motivation

- OmopSketch is designed to **summarise key information** of an OMOP-mapped database.
- Two main use cases:
 - **General characterisation** — understand what's in your database
 - **Feasibility assessment** — is this database suitable for my study?
- Available on [CRAN](#) and [GitHub](#).



Setup

Load packages and create the CDM object

- Load the required packages

```
library(OmopSketch)
library(CDMConnector)
library(duckdb)
library(dplyr)
library(here)
library(omock)
```

- Create a CDM object from the **GiBleed** synthetic dataset

```
con <- dbConnect(drv = duckdb(dbdir = here("Databases", "GiBleed.duckdb")))
```

```
[1] 0
```

```
cdm <- cdmFromCon(
  con = con,
  cdmSchema = "main",
  writeSchema = "results",
  writePrefix = "os_"
)
```

- The **cdm** object is your gateway to all OmopSketch functions – every function takes **cdm** as its first argument.

1. Snapshot

What is a snapshot?

`summariseOmapSnapshot()` captures **metadata about the CDM at a point in time** — useful for reproducibility and maintaining clear records of changes.

Information provided:

- **Vocabulary version** — from the `vocabulary` table
- **Table sizes** — record counts for `person` and `observation_period`
- **Observation period span** — earliest and latest dates in the data
- **Source type** — origin or type of the data source

summariseOmapSnapshot()

- Returns a **summarised result** object.

```
snapshot <- summariseOmapSnapshot(cdm = cdm)
```

```
glimpse(snapshot)
```

```
Rows: 16
Columns: 13
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
$ cdm_name       <chr> "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synt...
$ group_name     <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ group_level    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ variable_name  <chr> "general", "general", "general", "cdm", "cdm", "cdm", "cdm", "cdm", "cdm", "cdm", "observation_perio...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ estimate_name  <chr> "snapshot_date", "person_count", "vocabulary_version", "source_name", "version", "holder_name...
$ estimate_type  <chr> "date", "integer", "character", "character", "character", "character", "character", "character", "characte...
$ estimate_value <chr> "2026-09-17", "2694", "v5.0 18-JAN-19", "Synthea synthetic health database", "5.3", "OHDSI Co...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
```

tableOmopSnapshot ()

- Format the result into a **gt** table

```
tableOmopSnapshot(result = snapshot, type = "gt")
```

Snapshot of the cdm Synthea

Database name	
Synthea	
Estimate	
General	
Snapshot date	2026-09-17
Person count	2,694
Vocabulary version	v5.0 18-JAN-19
Cdm	
Source name	Synthea synthetic health database
Version	5.3
Holder name	OHDSI Community
Release date	2019-05-25
Description	SyntheaTM is a Synthetic Patient Population Simulator. The goal is to output synthetic, realistic (but not real), patient data and associated health records in a variety of formats.
Documentation reference	https://synthetichealth.github.io/synthea/
Observation period	
N	2,694

2. Observation period

What does the observation period tell us?

`summariseObservationPeriod()` summarises the **observation_period** table.

Information provided:

- **Number of records per person** — how many observation periods per individual
- **Duration in days** — length of each observation period
- **Days to next observation period** — gaps between consecutive periods
- **Data quality** (optional) — flags inconsistencies
- **Missingness** (optional) — counts of missing values per column

summariseObservationPeriod()

```
obs_result <- summariseObservationPeriod(cdm, missingData = TRUE, quality = TRUE)
glimpse(obs_result)
```

Rows: 3,126

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ cdm_name       <chr> "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synt...
$ group_name     <chr> "observation_period_ordinal", "observation_period_ordinal", "observation_period_ordinal", "ob...
$ group_level    <chr> "all", "all", "all", "all", "all", "all", "all", "all", "all", "all", "all", "all", "all", "all", "a...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ variable_name  <chr> "Records per person", "Records per person", "Records per person", "Records per person", "Reco...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ estimate_name  <chr> "mean", "sd", "min", "q05", "q25", "median", "q75", "q95", "max", "mean", "sd", "min", "q05",...
$ estimate_type  <chr> "numeric", "numeric", "integer", "integer", "integer", "integer", "integer", "integer", "inte...
$ estimate_value <chr> "1", "0", "1", "1", "1", "1", "1", "1", "1", "1", "14402.0014972862", "8725.3408283113", "1", "143...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
```


tableObservationPeriod()

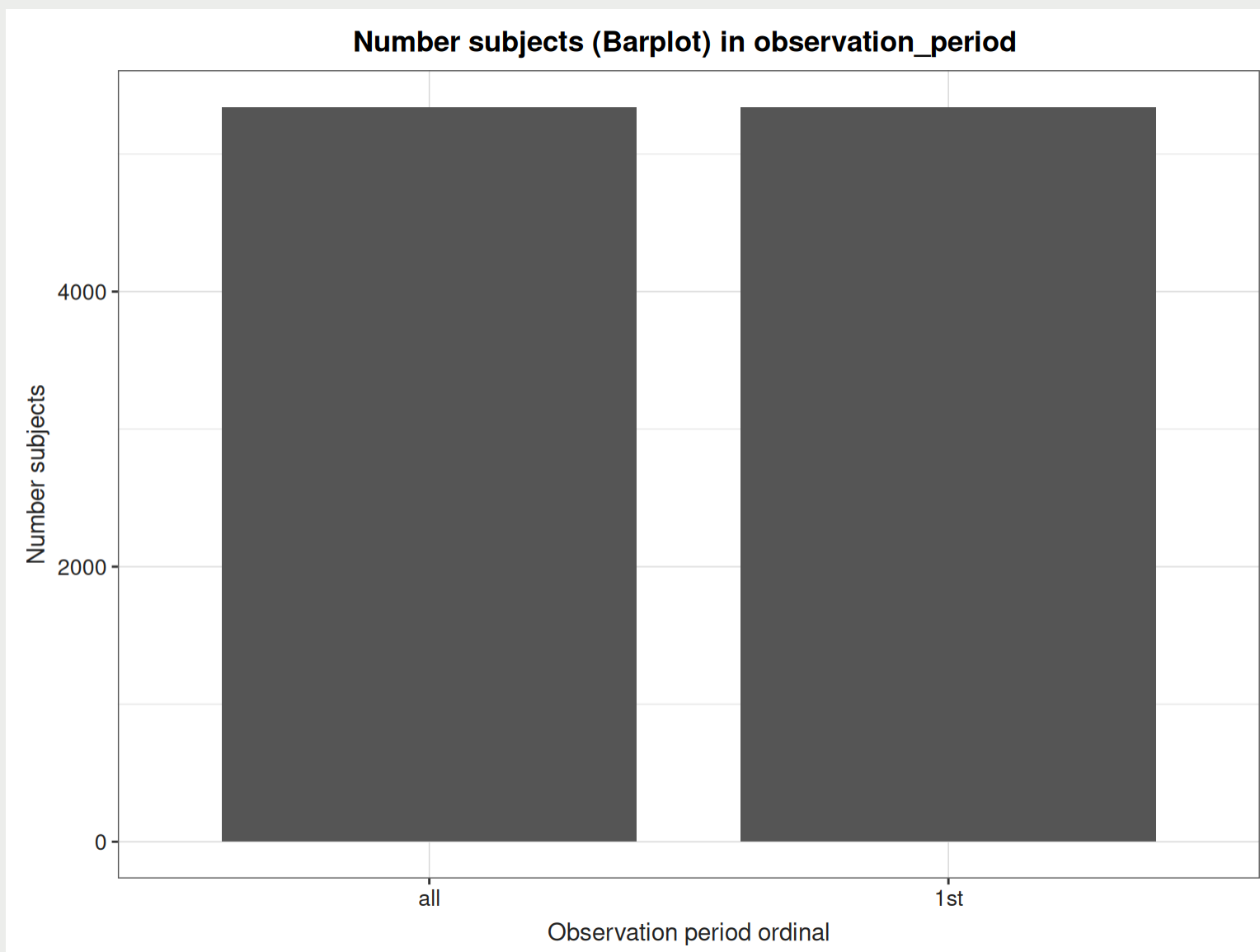
```
tableObservationPeriod(result = obs_result, type = "gt")
```

Summary of observation_period table

Observation period ordinal	Variable name	Variable level	Is required	Estimate name	Data source
					Synthea
all	Number records	–	overall	N	5,343
	Number subjects	–	overall	N	5,343
	Subjects not in person table	–	overall	N (%)	2,649 (49.58%)
	Records per person	–	overall	Mean (SD)	1.00 (0.00)
				Median [Q25 - Q75]	1 [1 - 1]
				Range [min to max]	[1 to 1]
	Duration in days	–	overall	Mean (SD)	14,402.00 (8,725.34)
				Median [Q25 - Q75]	13,868.00 [7,102.50 - 20,955.50]
				Range [min to max]	[1.00 to 40,349.00]
	Type concept id	Period covering healthcare encounters	overall	N (%)	5,343 (100.00%)
	Start date before birth date	–	overall	N (%)	0 (0.00%)
	End date before start date	–	overall	N (%)	0 (0.00%)
	Column name	Observation period end date	TRUE	N missing data (%)	0 (0.00%)

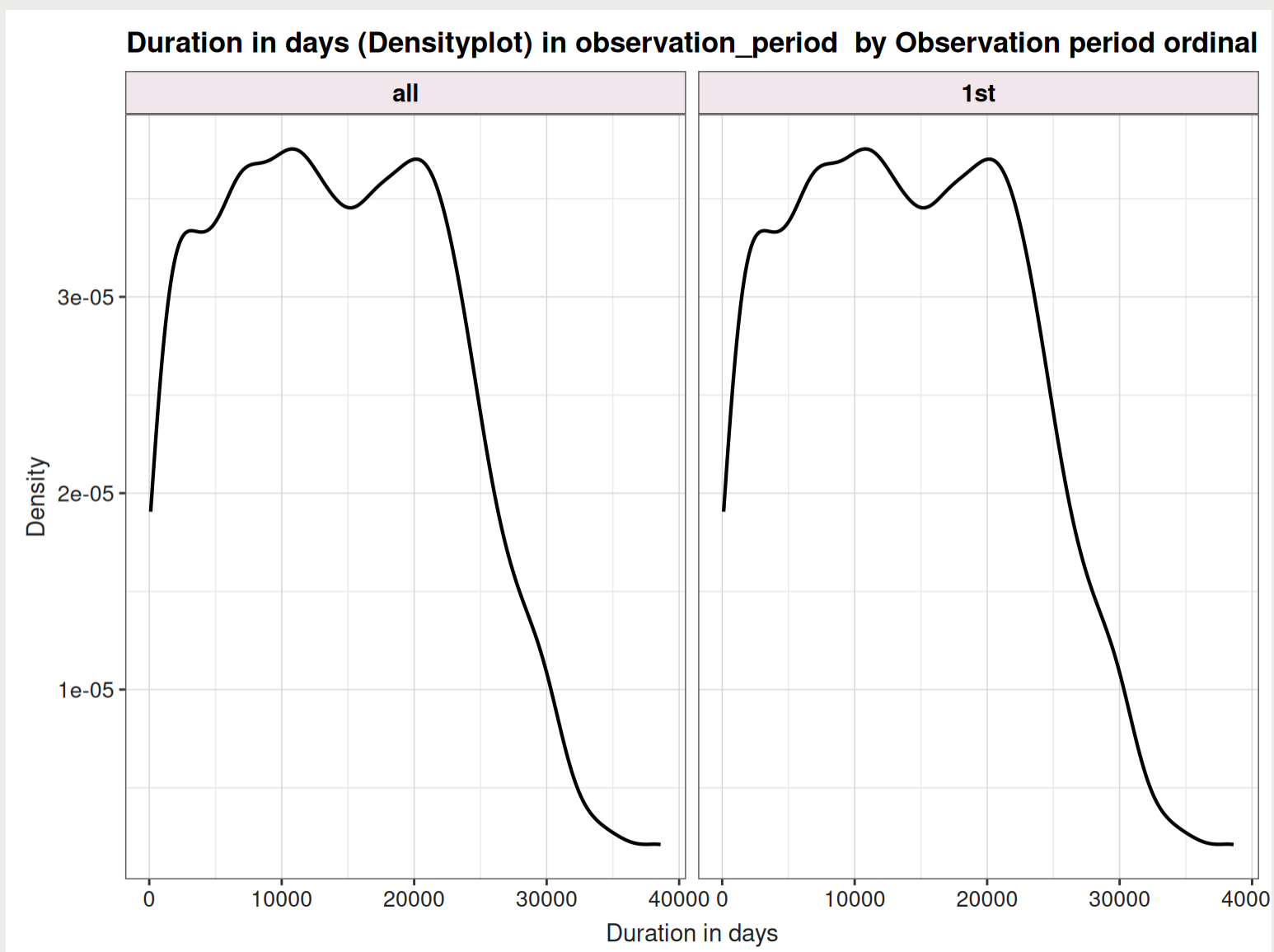
plotObservationPeriod() — Number of subjects

```
plotObservationPeriod(  
  result = obs_result,  
  plotType = "barplot",  
  variableName = "Number subjects"  
)
```



plotObservationPeriod() — Duration in days

```
plotObservationPeriod(  
  result = obs_result,  
  plotType = "densityplot",  
  variableName = "Duration in days",  
  facet = "observation_period_ordinal"  
)
```



3. Clinical tables

What does clinical table characterisation do?

`summariseClinicalRecords()` provides a **comprehensive audit** of any clinical table in the CDM.

Information provided:

summariseClinicalRecords()

```
clinical_result <- summariseClinicalRecords(  
  cdm          = cdm,  
  omopTableName = "condition_occurrence",  
  conceptSummary = TRUE,  
  missingData   = TRUE,  
  quality       = TRUE  
)
```

- You can pass multiple tables at once: `omopTableName = c("condition_occurrence", "drug_exposure")`

tableClinicalRecords()

```
tableClinicalRecords(result = clinical_result, type = "gt")
```

Summary of condition_occurrence table

Variable name	Variable level	Estimate name	Database name
			Synthea
condition_occurrence			
Number records	–	N	65,332
Number subjects	–	N (%)	2,694 (100.00%)
Subjects not in person table	–	N (%)	0 (0.00%)
Records per person	–	Mean (SD)	24.25 (7.41)
		Median [Q25 - Q75]	23 [19 - 29]
		Range [min to max]	[5 to 65]
Duration of records	–	Mean (SD)	53.17 (423.43)
		Median [Q25 - Q75]	14.00 [7.00 - 21.00]
		Range [min to max]	[0.00 to 18,979.00]
In observation	No	N (%)	450 (0.69%)
	Yes	N (%)	64,882 (99.31%)
Domain	Condition	N (%)	65,332 (100.00%)
Standard vocabulary	Snomed	N (%)	65,332 (100.00%)



Your turn: Clinical table audit

Task: Characterise **both** `drug_exposure` and `procedure_occurrence` in one call. Include missing data and quality checks.

Then answer:

1. Which table has more records per person on average?
2. What % of drug records are within the observation period?

►  [Click to see solution](#)

4. Trends over time

What do trends tell us?

- `summariseTrend()` reveals **how database content changes over time** — essential for spotting data gaps, coverage periods, and temporal changes.

event vs episode

`summariseTrend()` has two separate arguments for tables — and the choice changes **how records are counted across time intervals**.

event — tables where occurrences happen at a **single point in time**

- Each record is assigned to the interval containing its **start date only**
- e.g. a diagnosis on 3 March 2015 → counts in 2015, nowhere else
- Typical tables: `condition_occurrence`, `drug_exposure`, `procedure_occurrence`, `measurement`, `observation`, `visit_occurrence`, `death`

episode — tables that describe **periods spanning over time**

- Each record contributes to **every interval between its start and end date**
- e.g. an observation period from Jan 2015 to Dec 2016 → counts in both 2015 and 2016
- Typical tables: `observation_period`, `drug_era`, `condition_era`

Outputs available

output argument	What it counts
"record"	Number of records (default)
"person"	Number of distinct persons
"person-days"	Number of person-days (episode only)
"sex"	Number of females
"age"	Median age at start of period

summariseTrend()

```
trend_result <- summariseTrend(
  cdm      = cdm,
  event    = c("condition_occurrence", "drug_exposure"),
  interval = "years", # "years" | "quarters" | "months" | "overall"
  output   = "record", # "record" | "person" | "person-days" | "sex" | "age"
  dateRange = as.Date(c("1950-01-01", "2000-12-31"))
)

glimpse(trend_result)
```

Rows: 208

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ cdm_name       <chr> "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synt...
$ group_name     <chr> "omop_table", "omop_table", "omop_table", "omop_table", "omop_table", "omop_table", "omop_table", "omop_tab...
$ group_level    <chr> "condition_occurrence", "condition_occurrence", "drug_exposure", "drug_exposure", "condition_...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ variable_name  <chr> "Number of records", "Number of records", "Number of records", "Number of records", "Number o...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ estimate_name  <chr> "count", "percentage", "count", "percentage", "count", "percentage", "count", "percentage", "...
$ estimate_type  <chr> "integer", "percentage", "integer", "percentage", "integer", "percentage", "integer", "percen...
$ estimate_value <chr> "332", "0.77", "267", "0.61", "334", "0.78", "248", "0.57", "425", "0.99", "327", "0.75", "40...
$ additional_name <chr> "time_interval", "time_interval", "time_interval", "time_interval", "time_interval", "time_in...
```

summariseTrend() — check event/episode type

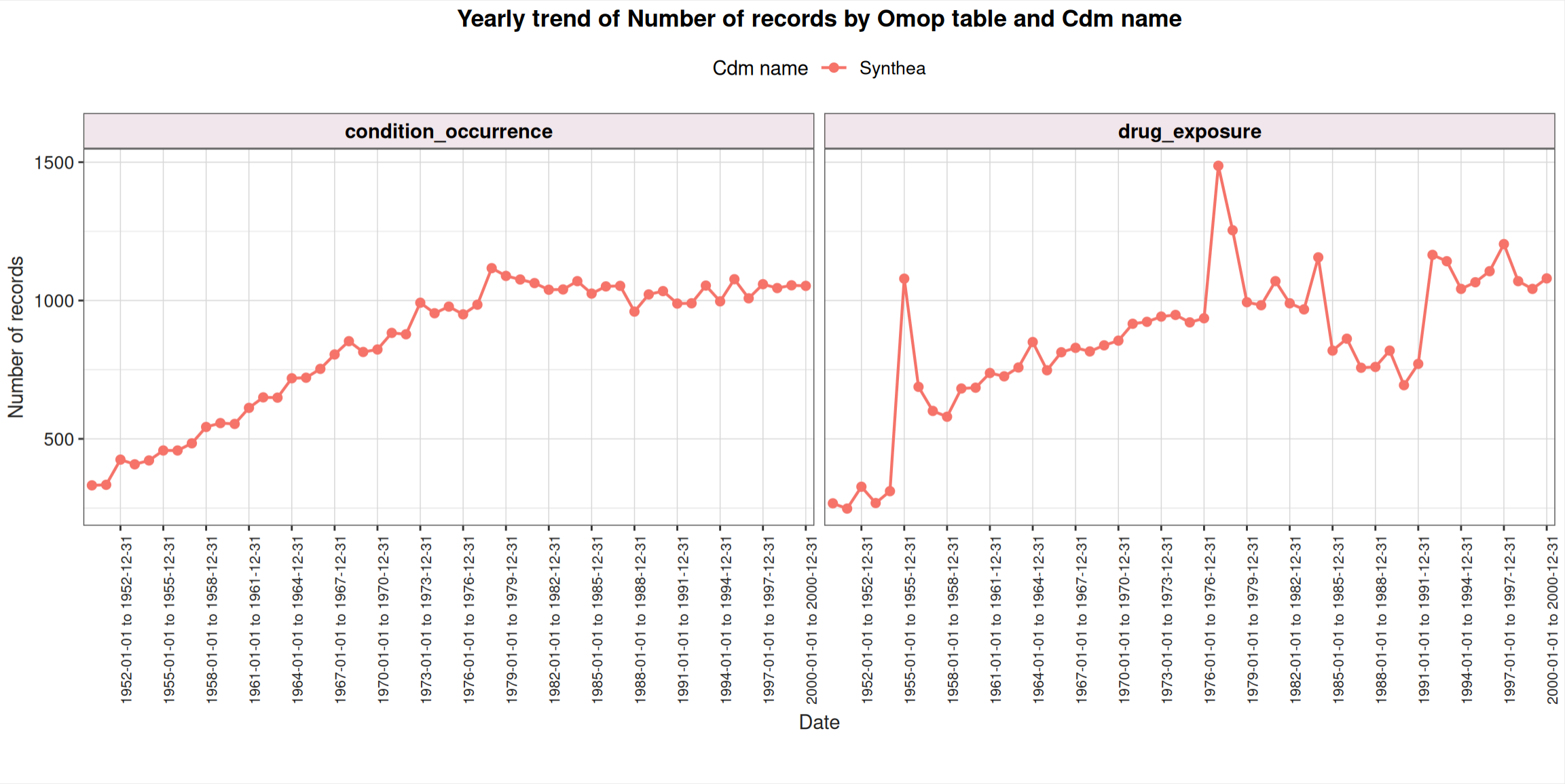
You can check how each table was treated in the result settings. The **type** column will show **"event"** or **"episode"** per table.

```
settings(trend_result) |> glimpse()
```

```
Rows: 1
Columns: 12
$ result_id      <int> 1
$ result_type    <chr> "summarise_trend"
$ package_name   <chr> "OmopSketch"
$ package_version <chr> "1.1.1"
$ group          <chr> "omop_table"
$ strata         <chr> ""
$ additional     <chr> "time_interval"
$ min_cell_count <chr> "0"
$ interval       <chr> "years"
$ study_period_end <chr> "2000-12-31"
$ study_period_start <chr> "1950-01-01"
$ type          <chr> "event"
```

plotTrend()

```
plotTrend(result = trend_result, facet = "omop_table", colour = "cdm_name")
```



tableTrend()

```
tableTrend(result = trend_result, type = "reactable")
```

Search							Database name
Type; OMOP table	Variable name	Time interval	Estimate name	Interval	Study period end	Study period start	Synthea
▼event; condition _occurre							



Your turn: Sex-stratified trends

Task: Create a plot showing the **yearly** trend in records for **drug_exposure** and **condition_occurrence**, stratified by **sex**, for the date range **1990–2010**.

Hints:

- `output = "record"` gives you record counts
- `sex = TRUE` gives you sex-stratified summaries
- `dateRange = as.Date(c("1990-01-01", "2010-12-31"))`
- Use `facet = "omop_table"` and `colour = "sex"` in the plot

►  [Click to see solution](#)

5. Concept counts

summariseConceptIdCounts()

- Counts how often each concept ID appears — useful for **feasibility** (do we have enough patients with condition X?).

```
concept_result <- summariseConceptIdCounts(
  cdm          = cdm,
  omopTableName = "procedure_occurrence",
  countBy      = c("record", "person")
)
glimpse(concept_result)
```

Rows: 102

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ cdm_name       <chr> "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synthea", "Synt...
$ group_name     <chr> "omop_table", "omop_table", "omop_table", "omop_table", "omop_table", "omop_table", "omop_table", "omop_tab...
$ group_level    <chr> "procedure_occurrence", "procedure_occurrence", "procedure_occurrence", "procedure_occurrence...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ variable_name  <chr> "Suture open wound", "Suture open wound", "Direct current cardioversion", "Direct current car...
$ variable_level <chr> "4125906", "4125906", "4078793", "4078793", "4163872", "4163872", "4037675", "4037675", "4181...
$ estimate_name  <chr> "count_records", "count_subjects", "count_records", "count_subjects", "count_records", "count...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "inte...
$ estimate_value <chr> "2487", "1606", "1342", "134", "652", "574", "102", "102", "133", "79", "138", "138", "908", ...
$ additional_name <chr> "source_concept_id &&& source_concept_name &&& standard_vocabulary &&& source_vocabulary", "s...
```

tableConceptIdCounts()

- Filter using the **display** argument:

display	What is shown
"overall" (default)	All records
"standard"	Standard concepts only
"source"	Source codes only
"missing standard"	Source codes with no standard mapping
"missing source"	Standard concepts with no source code

tableConceptIdCounts()

```
tableConceptIdCounts(result = concept_result, display = "standard", type = "reactable")
```

Search						
Database	OMOP table	Standard concept name	Standard concept id	Standard vocabulary	N records	N subjects
Synthea	procedure_occurrence	Allergy screening test	4191853	Snomed	704	320
Synthea	procedure_occurrence	Appendectomy	4198190	Snomed	157	157
Synthea	procedure_occurrence	Asthma screening	4062501	Snomed	101	101

ce						
Synthea	procedure _occurrence	Bilateral tubal ligation	4117038	Snomed	12	12
Synthea	procedure _occurrence	Bone density scan	4195803	Snomed	655	491
Synthea	procedure _occurrence	Bone immobiliz ation	4170947	Snomed	1789	1278
Synthea	procedure _occurrence	Brief general examinati on	4037675	Snomed	102	102
Synthea	procedure _occurrence	Catheter ablation of	4057420	Snomed	438	236

← Back to Presentations

		tissue of heart				
Synthea	procedure _occurrence	Cholecyste ctomy	4242997	Snomed	3	3
Synthea	procedure _occurrence	Cognitive and behavioral therapy	4043071	Snomed	2820	146
1–10 of 51 rows		Show	10 ▾	Previous 1 2 3 4 5 6 Next		

tableTopConceptCounts()

```
tableTopConceptCounts(result = concept_result, countBy = "record", top = 10, type = "gt")
```

Top 10 concepts in procedure_occurrence table ranked by record count

Top	Data source
	Synthea
procedure_occurrence	
1	Standard: Subcutaneous immunotherapy (4107731 - SNOMED) Source: Subcutaneous immunotherapy (4107731 - SNOMED) 17520
2	Standard: Cognitive and behavioral therapy (4043071 - SNOMED) Source: Cognitive and behavioral therapy (4043071 - SNOMED) 2820
3	Standard: Suture open wound (4125906 - SNOMED) Source: Suture open wound (4125906 - SNOMED) 2487
4	Standard: Bone immobilization (4170947 - SNOMED) Source: Bone immobilization (4170947 - SNOMED) 1789
5	Standard: Sputum examination (4151422 - SNOMED) Source: Sputum examination (4151422 - SNOMED) 1659
6	Standard: Direct current cardioversion (4078793 - SNOMED) Source: Direct current cardioversion (4078793 - SNOMED) 1342
7	Standard: Pulmonary rehabilitation (4035793 - SNOMED) Source: Pulmonary rehabilitation (4035793 - SNOMED) 908
8	Standard: Intramuscular injection (4295880 - SNOMED) Source: Intramuscular injection (4295880 - SNOMED) 908
9	Standard: Allergy screening test (4191853 - SNOMED) Source: Allergy screening test (4191853 - SNOMED)

Top	Data source
	Synthea
	704
10	Standard: Bone density scan (4195803 - SNOMED) Source: Bone density scan (4195803 - SNOMED) 655

[← Back to Presentations](#)

6. Full database characterisation

databaseCharacteristics()

- `databaseCharacteristics()` runs **all summaries in one call**.

```
full_result <- databaseCharacteristics(  
  cdm                = cdm,  
  omopTableName      = c("drug_exposure", "condition_occurrence"),  
  sex                = TRUE,  
  ageGroup           = list(c(0, 50), c(51, Inf)),  
  interval           = "years",  
  dateRange          = as.Date(c("2010-01-01", "2020-12-31")),  
  conceptIdCounts    = FALSE  
)
```

shinyCharacteristics()

- Then launch a **Shiny app** to explore everything interactively:

```
shinyCharacteristics(result = full_result, directory = here())
```

Live Shiny demo

 Database characterisation

 Background

 Snapshot

 Population Characteristics

 Person Table Summary

 Observation Period

 Clinical Tables

 Concept Counts

OmopSketch Database Characterisation

Shiny App Overview

This Shiny App presents the results of a database characterisation performed using the **OmopSketch** R package. It provides summaries and visualisations to support exploration of OMOP CDM-compliant datasets.

The app includes the following components:

- **Snapshot:** Metadata extracted from the `cdm_source` table, using the output of `summariseOmopSnapshot()`.
- **Population Characteristics:** Summary of the demographics of the population in observation, generated using **CohortConstructor** and **CohortCharacteristics**.
- **Person:** Summary of person table, from `summarisePerson()`.
- **Observation Period:** Distribution and length of observation periods, based on `summariseObservationPeriod()`.
- **Trends:** Temporal trends of individuals and records in observation, including changes in median age, proportion of females, and number of person-days, generated from `summariseTrend()`.
- **Clinical Records:** Summary of clinical tables focused on vocabulary usage and quality checks, from `summariseClinicalRecords()`.
- **Concept Counts** (*optional*): Counts of `concept_id` s across tables, generated by `summariseConceptIdCounts()`.

Shiny generation

↗ Open full app

Resources

👉 Package website

👉 CRAN

👉 PDF manual

✉ cecilia.campanile@ndorms.ox.ac.uk